Monday, 23 March 2026 8:03 AM

Decision Tree

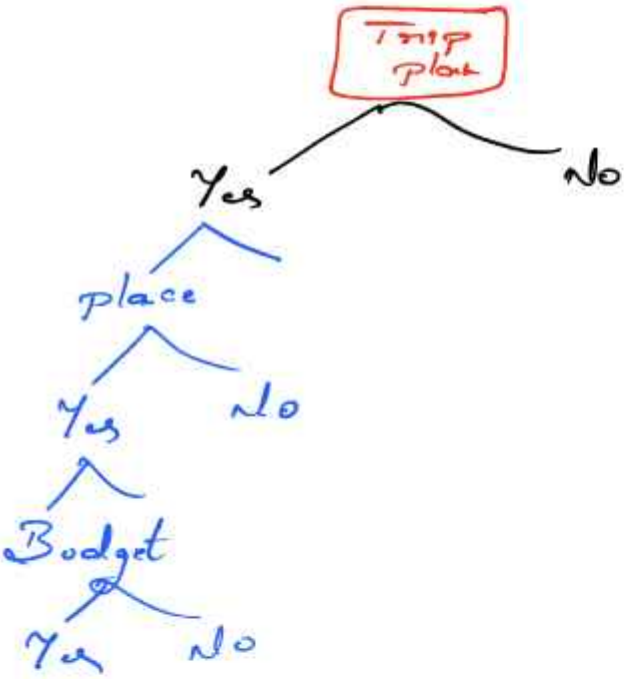
① It Comes under Supervised $\begin{cases} X \\ Y \end{cases}$

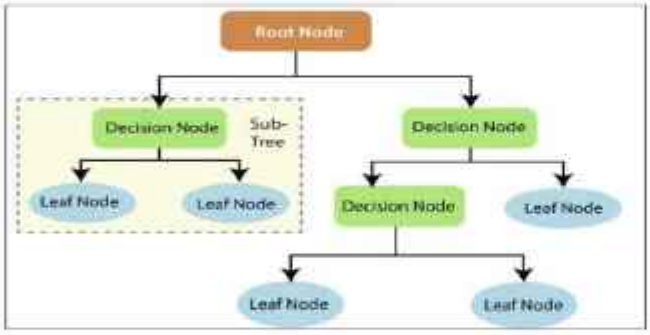
② Decision tree can be used to solve both regression as well as the classification

↳ Decision Tree Regression

↳ Decision Tree Classification.

Decision tree uses if-else logic to Construct Tree.





Root Node

Root node is main node which

represents entire population data which gets splitted into Homogeneous node.

Splitting

Process of dividing nodes into subnodes.

Decision node

Node at which decision making will happen by dividing nodes into further subnodes. Decision node will include leaf node.

Leaf Node

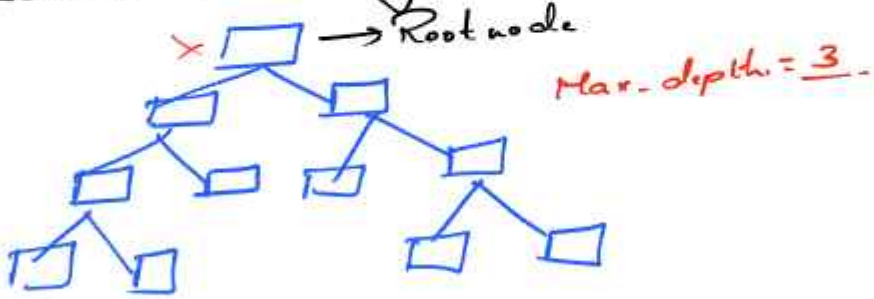
Leaf Nodes are end points where no further split happens. It contains final classification or prediction.

Sub-tree

Branch or part of tree with Decision nodes or leaf nodes.

Pruning

Process of removing nodes to reduce the height of a tree.

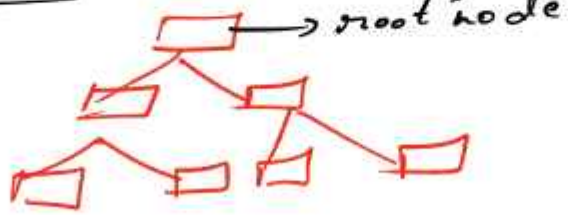


11 Decision Tree works?

How T ...

Decision is used to solve both Regression as well classification by Constructing tree structure.

Decision tree breaks down entire dataset into smaller and smaller chunks so that an incremental tree will be developed with Decision nodes and leaf nodes. The End result will be the Decision Tree



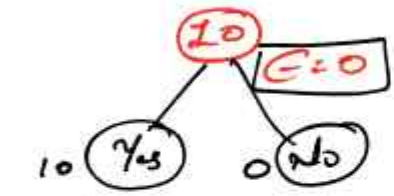
How to Select root Node?

When we have many Columns in a dataset, how do we select one Column as root node

→ Entropy

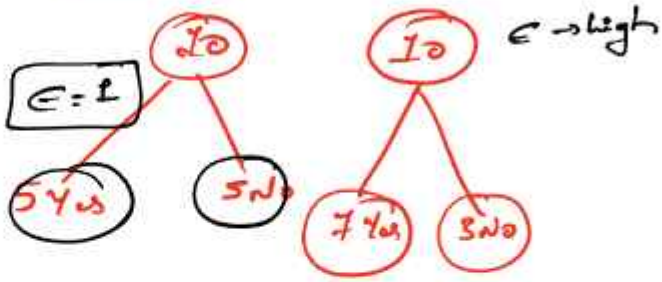
→ Gini Index

Pure Split



A node which Contains 100% of one class.

Impure split



Node Contains Mix of both classes.

Entropy

Entropy is a measure of randomness or uncertainity in a data.

Pure split → Entropy = 0

Impure split → Entropy will be high.

Entropy measures Impurity

$$H(S) = -P_+ \log_2 P_+ - P_- \log_2 P_-$$

P_+ → probability of positive class

P_- → Probability of negative class

Information Gain

① Information Gain measures how much information given by each Column.

② Root node will be the one which gives highest information Gain.

$$\text{Gain} = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$$

$H(S)$ → Entropy before split

S → Sample before split

$H(S_v)$ → Entropy after split

S_v → Sample after split

How it works?

① Find overall Entropy of Column before split

② Find Entropy for each category

③ Find information Gain

④ Repeat steps (1, 2, 3) for all Columns

⑤ Select the Column which has got highest information Gain.

x_1	x_2	x_3	y
Weather	Humidity	Wind	play

① Entropy before split

sunny	high	weak	no
sunny	high	strong	no
cloudy	high	weak	yes
rainy	high	weak	yes
rainy	normal	weak	yes
rainy	normal	strong	no
cloudy	normal	strong	yes
sunny	high	weak	no
sunny	normal	weak	yes
rainy	normal	weak	yes
sunny	normal	strong	yes
cloudy	high	strong	yes
cloudy	normal	weak	yes
rainy	high	strong	no

Play

9 Yes 5 No

$P(Yes) = \frac{9}{14}$

$P(No) = \frac{5}{14}$

$H(S) = -P_Y \log_2 P_Y - P_N \log_2 P_N$

$H(S) = -\left(\frac{9}{14}\right)\log_2\left(\frac{9}{14}\right) - \left(\frac{5}{14}\right)\log_2\left(\frac{5}{14}\right)$

$\boxed{H(S) = 0.92}$

② weather

Sunny Cloudy Rainy

Entropy of Sunny

① Sunny

2 Yes 3 No

$P(Y) = \frac{2}{5}$

$P(N) = \frac{3}{5}$

$H(Sunny) = -\left(\frac{2}{5}\right)\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)$

$\boxed{H(Sunny) = 0.97}$

Weather	Humidity	Wind	play
sunny	high	weak	no
sunny	high	strong	no
cloudy	high	weak	yes
rainy	high	weak	yes
rainy	normal	weak	yes
rainy	normal	strong	no
cloudy	normal	strong	yes
sunny	high	weak	no
sunny	normal	weak	yes
rainy	normal	weak	yes
sunny	normal	strong	yes
cloudy	high	strong	yes
cloudy	normal	weak	yes
rainy	high	strong	no

Entropy of Cloudy

Cloudy pure split

4 Yes 0 No

$P(Y) = 1$ $P(N) = 0$

$H(cloudy) = -1\log_2(1) - 0 = 0$

$\boxed{H(cloudy) = 0}$

Entropy of rainy

rainy

Weather	Humidity	Wind	play
sunny	high	weak	no
sunny	high	strong	no

cloudy	high	weak	yes
rainy	high	weak	yes
rainy	normal	weak	yes
rainy	normal	strong	no
cloudy	normal	strong	yes
sunny	high	weak	no
sunny	normal	weak	yes
rainy	normal	weak	yes
sunny	normal	strong	yes
cloudy	high	strong	yes
cloudy	normal	weak	yes
rainy	high	strong	no

3 Yes 2 No

$P(Y) = \frac{3}{5}$, $P(N) = \frac{2}{5}$

$$H(\text{rainy}) = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$$

$$\boxed{H(\text{rainy}) = 0.97}$$

Find Information Gain.

$$\text{Gain} = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$$

$$= H(S) - \frac{S_{sunny}}{S} H(\text{Sunny}) - \frac{S_{rainy}}{S} H(\text{rainy}) - \cancel{\frac{S_{cloudy}}{S} H(\text{cloudy})} \to 0$$

$$= 0.92 - \frac{5}{14}(0.97) - \frac{5}{14}(0.97)$$

$$\boxed{\text{Gain}_{\text{weather}} = 0.247}$$

lly

Gain(weather) = 0.247

Gain(Humidity) = 0.17

Gain(wind) = 0.15

Since information Gain is high for weather, root node is weather

Weather	Humidity	Wind	play
sunny	high	weak	no
sunny	high	strong	no
cloudy	high	weak	yes
rainy	high	weak	yes
rainy	normal	weak	yes
rainy	normal	strong	no
cloudy	normal	strong	yes
sunny	high	weak	no
sunny	normal	weak	yes
rainy	normal	weak	yes
sunny	normal	strong	yes
cloudy	high	strong	yes
cloudy	normal	weak	yes

rainy	high	strong	no

Limitation

① Computationally Expensive because of $\log_2$

↳ Gini Index, weighted Gini